

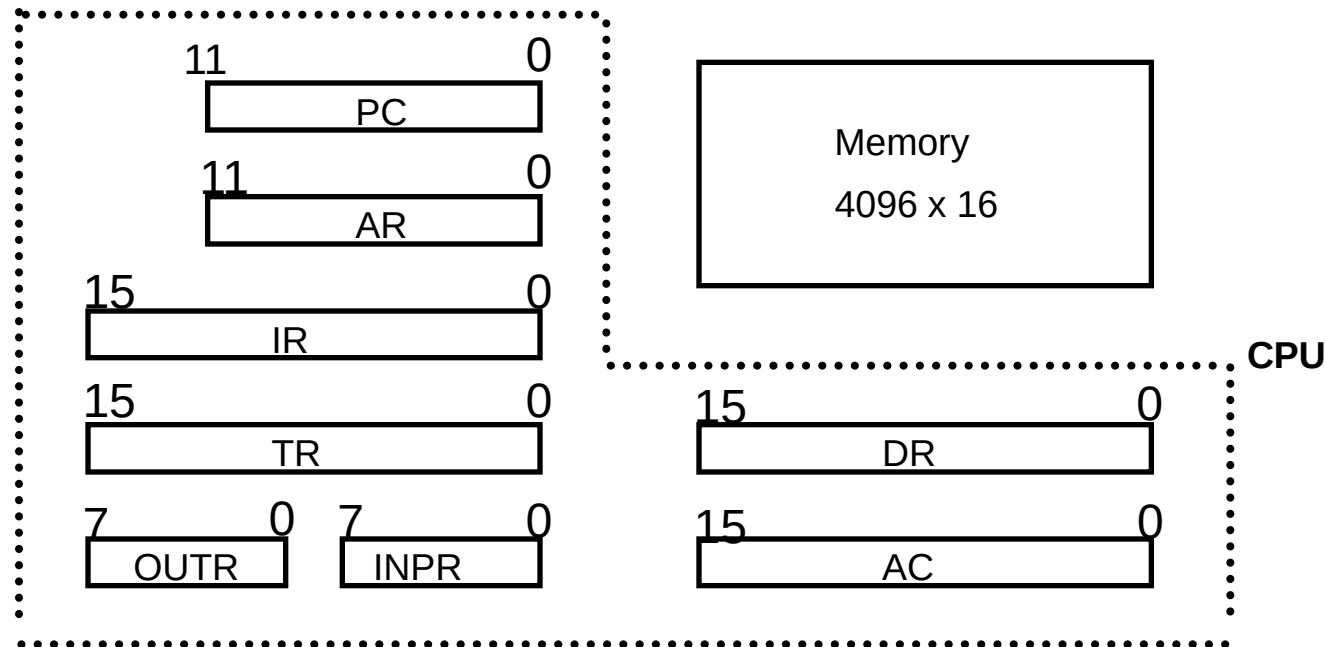
PROCESSOR REGISTERS

- The significance of a general purpose register is that it can be referred to in instructions
 - e.g. load AC with the contents of a specific memory location; store the contents of AC into a specified memory location
- Often a processor will need a scratch register to store intermediate results or other temporary data; in the Basic Computer this is the *Temporary Register* (TR)
- The Basic Computer uses a very simple model of input/output (I/O) operations
 - Input devices are considered to send 8 bits of character data to the processor
 - The processor can send 8 bits of character data to output devices
- The *Input Register* (INPR) holds an 8 bit character gotten from an input device
- The *Output Register* (OUTR) holds an 8 bit character to be send to an output device



BASIC COMPUTER REGISTERS

Registers in the Basic Computer



List of BC Registers

DR	16	Data Register	Holds memory operand
AR	12	Address Register	Holds address for memory
AC	16	Accumulator	Processor register
IR	16	Instruction Register	Holds instruction code
PC	12	Program Counter	Holds address of instruction
TR	16	Temporary Register	Holds temporary data
INPR	8	Input Register	Holds input character
OUTR	8	Output Register	Holds output character

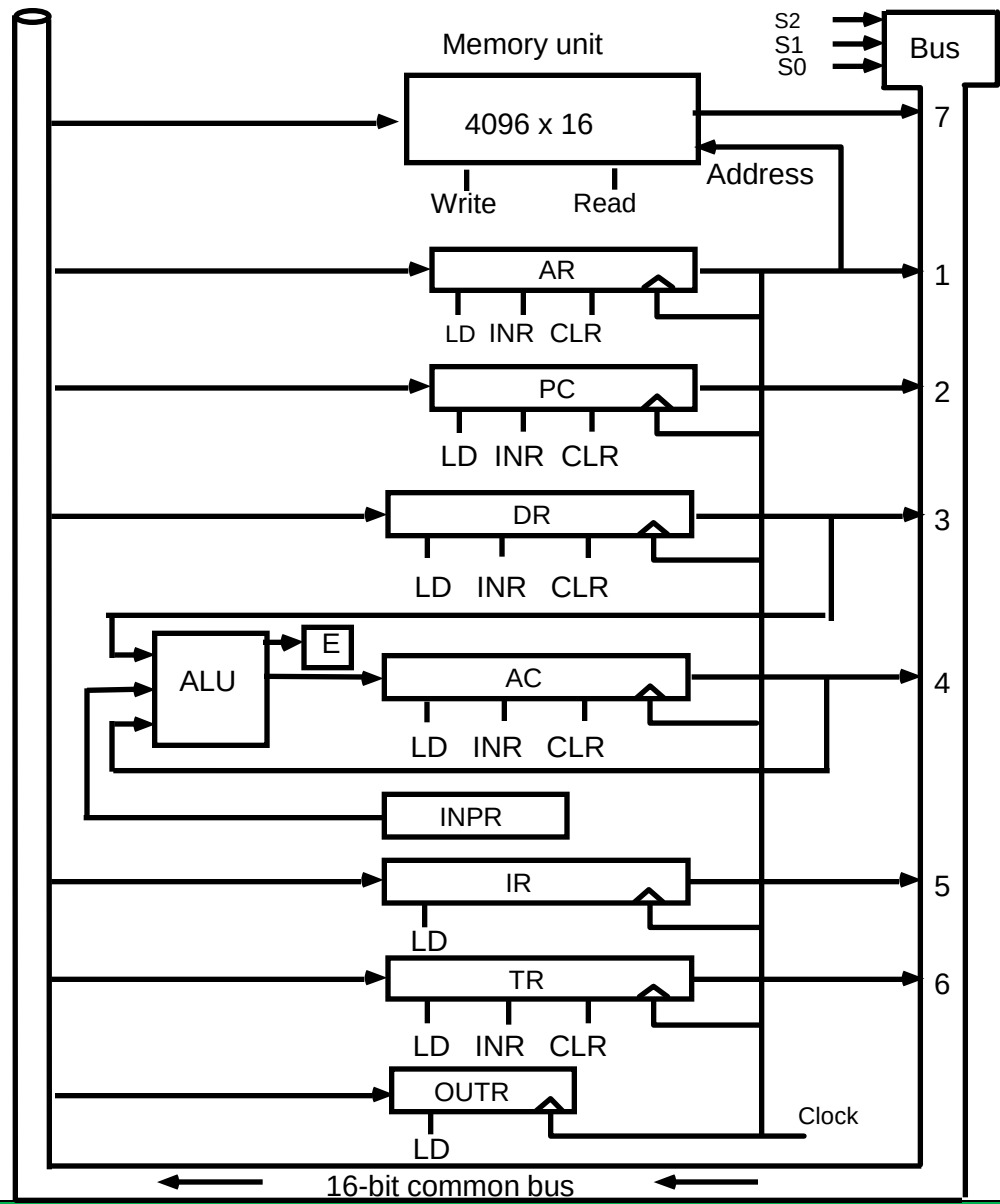


COMMON BUS SYSTEM

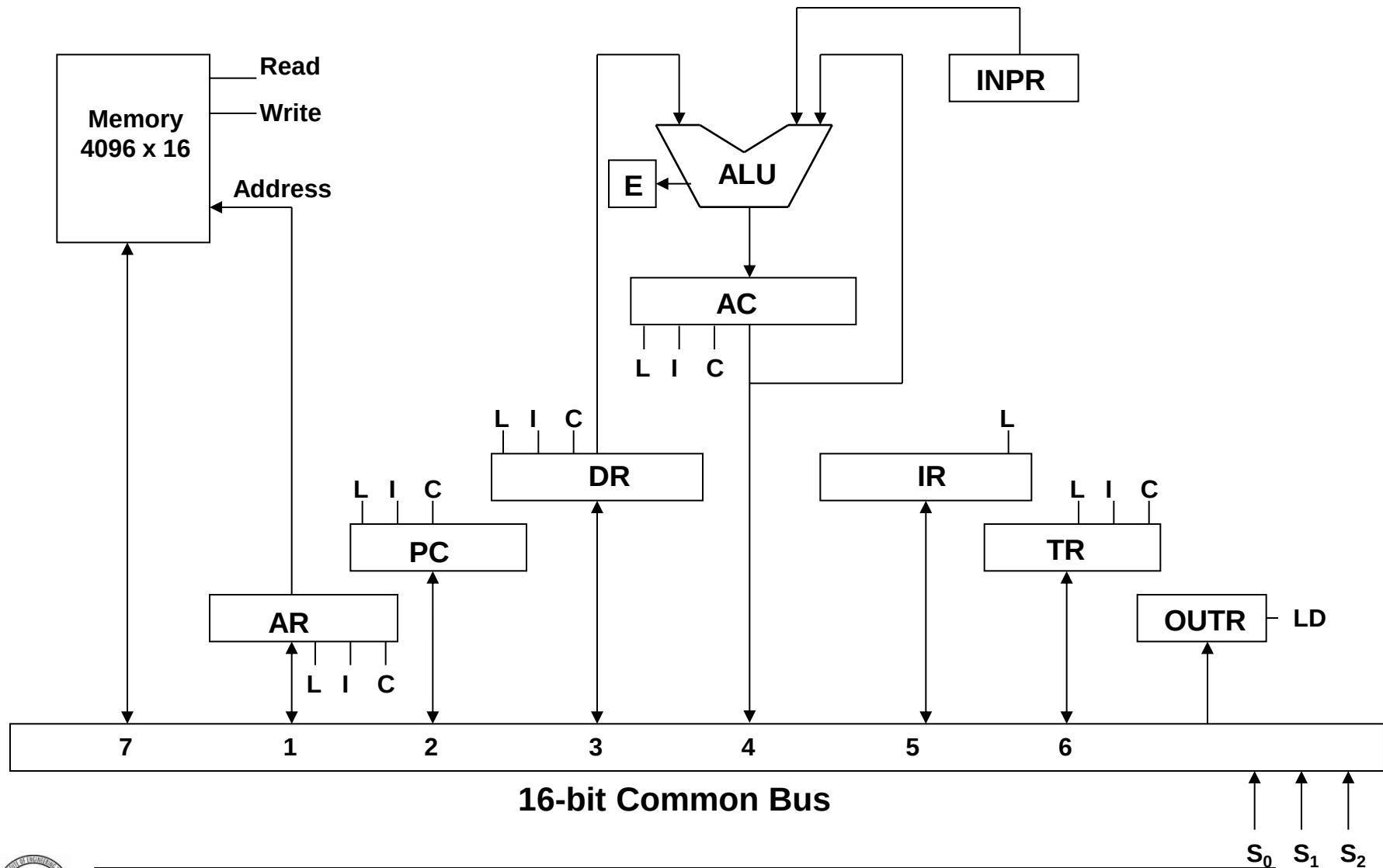
- The registers in the Basic Computer are connected using a bus
- This gives a savings in circuitry over complete connections between registers



COMMON BUS SYSTEM



COMMON BUS SYSTEM



COMMON BUS SYSTEM

- Three control lines, S_2 , S_1 , and S_0 control which register the bus selects as its input

S_2	S_1	S_0	Register
0	0	0	X
0	0	1	AR
0	1	0	PC
0	1	1	DR
1	0	0	AC
1	0	1	IR
1	1	0	TR
1	1	1	Memory

- Either one of the registers will have its load signal activated, or the memory will have its read signal activated
 - Will determine where the data from the bus gets loaded
- The 12-bit registers, AR and PC, have 0's loaded onto the bus in the high order 4 bit positions
- When the 8-bit register OUTF is loaded from the bus, the data comes from the low order 8 bits on the bus

